Team Awesome

Team 1234A

Engineering Notebook

Notebook Number: 1

Date: September 1, 2023 – April 30, 2024

Organization: Awesome High School Robotics

Team Members:

Alice Smith, Bob Jones, Charlie Brown

Design Process Phases & Formatting Guide

This notebook is color-coded by the phase of the engineering design process. The table of contents and the header of each entry will reflect these phases.

Define Problem: Identifying the core issue, setting SMART goals, outlining constraints and deliverables.

Generate Concepts: Brainstorming, research, prototyping, and decision matrices to evaluate potential solutions.

Develop Solution: Creating CAD, detailed sketches, math calculations, graphical models, and pseudocode.

Construct and Test: Building the robot, writing the code, executing test plans, and gathering qualitative/quantitative data.

Evaluate Solution: Reflecting on constraints, event outcomes, and planning future improvements.

General Formatting

- Keep entries in chronological order.
- Ensure all figures have captions and the initials of the author.
- Provide detailed explanations for calculations, code logic, and design choices.

Kickoff and Initial Brainstorming

Project Title: Kickoff and Initial Brainstorming **Date:** September 1, 2023
Author: Alice Smith **Phase:** Define Problem

Goals for the Meeting

Today we watched the game reveal and defined our initial goals and criteria.

Problem Statement

We need to build a robot that can reliably pick up objects and score them in the high goal, while also being able to traverse the terrain.

Initial Thoughts

We think a flywheel might be a good idea. We should research different types of intakes.

Table 1: Initial Design Criteria

Criteria	Priority
Shoot from afar	High
Fast drive	Medium
Climb	Low

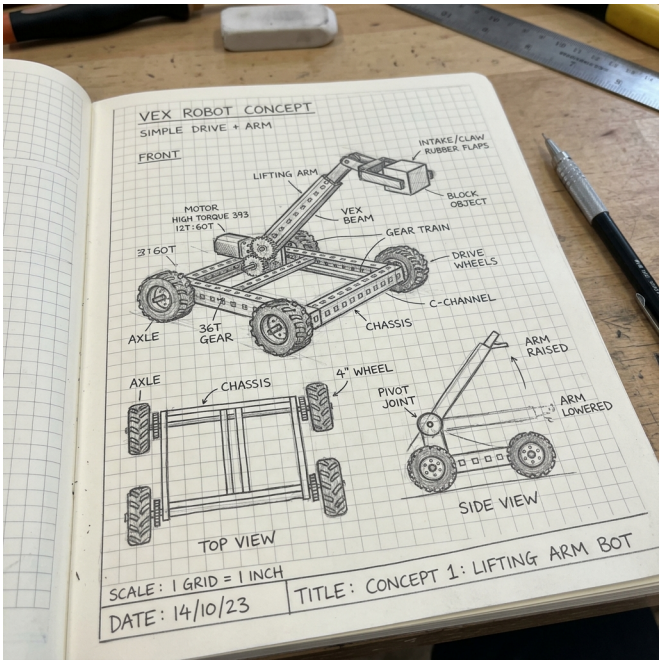

Figure 1: Sketch of our first concept (Initials: AS)

Next Steps

Next meeting we will begin brainstorming specific mechanisms and generating concept matrices.